Flex & Rigid-Flex PCBs

Bendable circuits for wearables, automotive & space-constrained products

Polyimide • Stiffener • Bend radius • Static vs Dynamic

Audience: Intermediate Engineering Students

Why • Materials • Layout rules • DFM • Reliability

Why Flex / Rigid-Flex?

When a rigid board just won't fit

Tight 3-D fitment

Wraps around a battery,
lens, or chassis.

Weight reduction

No connectors, less material —
lighter assemblies.

Connector elimination

Direct interconnect →
fewer failure points.

Dynamic flexing

Camera tilt, hinge, foldable
phones, AR/VR.

Vibration resistance

Solid integral structure —
automotive, aerospace.

Thinner products

< 1 mm total thickness
for wearables & ID-1 cards.

Flex Circuit Variants

Single-sided / double-sided / multilayer / rigid-flex

Type	Conductors	Use	Cost
1-layer flex	Copper one side only	LED strips, simple jumpers	Low
2-layer flex	Plated through-holes	Sensor-to-MCU interconnect	Moderate
Multilayer flex	3-12 layers	Dense compact assemblies	High
Rigid-Flex	Rigid sections + flex tail	Phones, cameras, drones	High
Stiffener flex	Rigid backers added selectively	Connector mounting areas	Moderate
Flex Heat-Set	Pre-formed into 3D shape	Aerospace, medical implants	Very high

Flex Materials Stack

Polyimide + adhesive + copper — repeated

- Polyimide (Kapton) — base film, ~25-50 µm thick
- Adhesive layer — acrylic or epoxy, bonds copper to polyimide
- Rolled-Annealed (RA) copper — ductile, survives bending
- Electrodeposited (ED) copper — cheaper but more brittle
- Coverlay — polyimide film replacing solder mask
- Stiffener — FR-4 / steel / polyimide pieces for rigid mount points
- ZIF/FFC connectors expect specific thickness (0.3 mm common)

Bend Radius Rules

The single most important flex design parameter

Application	Bend radius (min)	Notes
Single-sided flex, static	6 × thickness	One-time forming
Double-sided flex, static	10 × thickness	After assembly
Multilayer flex, static	15 × thickness	Avoid sharp bends
Dynamic flex (camera, hinge)	100 × thickness	Survives ≥ 100k cycles
Continuous dynamic flex	200 × thickness	Robot arms, special apps

What happens past the limit

Copper cracks in the bend zone → open circuits. Worse: sometimes the failure is intermittent and only appears with vibration.

Flex Layout Rules

Choices that determine reliability under bending

- Traces perpendicular to bend axis — they take the least stress
- Avoid via-in-bend regions
- Stagger traces on different layers (no overlap in the bend area)
- Tear-drop pad junctions where traces meet pads
- Larger pad rings than on rigid PCBs
- Use rolled-annealed copper for dynamic flex
- Cross-hatched ground plane (not solid) — solid copper too stiff to bend
- Add 'service loops' — extra length so the cable can move freely

Flex Connectors

How flex meets the rest of the assembly

FFC / FPC

Flat-flex with ZIF or LIF
connector on rigid PCB.

Solder pads

Direct solder of flex tail
to rigid PCB pads.

Hot-bar bonding

Anisotropic conductive film
(ACF) heat-pressed.

LCP / Polyimide

Higher-temp variants
for RF and exotic apps.

FFC sizes

1.0 / 0.5 / 0.3 mm pitch.
0.5 mm dominates consumer.

Stiffener under

Adds rigidity for
connector insertion force.

Rigid-Flex Construction

Rigid islands joined by flex tails

- Rigid sections built from standard FR-4
- Flex layers continuous through rigid sections (sub-laminated)
- Eliminates cables / connectors between sub-boards
- Higher reliability — fewer mechanical joints
- Cost: ~3-5× standard rigid PCB
- Used in: cameras, drones, foldable phones, satellites, medical implants
- Design tools: Altium, Allegro, KiCad 8+ (basic), Cadence Sigrity

DFM for Flex

Get the fab to agree with you early

- Talk to the fab BEFORE designing — stackup matters
- Ask for capability sheet: trace width, drill, layer count
- Coverlay openings need correct tolerance
- Specify stiffener locations precisely
- Tear-drops and fillets at trace-pad junctions
- Avoid 90° trace corners — use curves
- Service loops in cables
- Annotate dynamic vs static bend zones
- Test points on rigid sections only
- Marker layer for fold lines in documentation
- Plan for assembly fixture (how it's positioned)
- Consider thickness of finished assembly with stiffener

Reliability & Testing

Cycle life, environmental

Cycle test

Bend rig measures cycles
until resistance change.

Thermal cycle

-40 to +85 °C × 1000 cycles
for automotive use.

Humidity

85 °C / 85% RH bias
then retest electrical.

Salt fog

1000 hrs for marine
or outdoor products.

Vibration

Random or sine.
For automotive / aerospace.

Drop test

Phones / wearables.
1 m drop on hard floor.

Cost — Flex vs Rigid Compare

Order of magnitude pricing for protos

Type	Proto (5 pcs)	Volume (10k)	Lead time
2-layer rigid FR-4	₹500-2k	₹50-100 each	5-10 days
2-layer flex	₹3-8k	₹150-300	10-20 days
Multilayer rigid (4)	₹2-5k	₹200-400	5-10 days
Multilayer flex (4)	₹10-20k	₹500-1500	20-30 days
Rigid-flex (8 layer)	₹40-80k	₹2k-5k	30-45 days

Real-World Applications

Where flex shows up in products you know

Smartphone hinge

Flex between display and main board.

Camera modules

Imaging sensor → main board over tight 3D path.

Wearables

Band, sensors, battery — all on one flex circuit.

Drones

Gimbal motors, FPV camera, GPS.

Automotive

Dashboard, mirrors, steering wheel.

Medical implants

Pacemakers, neurostim — flex for safety & geometry.

Common Flex Mistakes

What 'send to fab' reveals

- Bend radius too small — copper cracks during assembly or in service
- Vias placed in the flex bend area
- Solid copper plane on flex — too stiff to bend, causes failure
- Sharp 90° trace corners — concentrate stress
- No service loop — flex stretches, breaks at solder joint
- Stiffener missing where connector inserts
- Wrong coverlay opening tolerance — pads exposed or covered
- Forgotten: order the flex from a flex-specialist fab, not your regular rigid-PCB shop

Key Takeaways

Key takeaways from this lecture

Right tool for 3D fitment

Flex when rigid doesn't fit.

Bend radius is sacred

Dynamic 100× t, static 10× t.

Traces perpendicular to bend

Lower copper stress.

Hatched plane on flex

Solid is too stiff.

Specialist fab needed

Don't use regular rigid shop.

Cost 3-5× rigid

Plan budget early.

Recommended Reading

Flex design resources

- IPC-2223 — Sectional Design Standard for Flexible Printed Boards
- DuPont Pyralux design guide
- Multech / All Flex / FlexHub / Streamline Circuits — fab vendor guides
- Phil's Lab YouTube — flex PCB walkthrough
- Sierra Circuits — free flex design e-book
- Altium / KiCad rigid-flex tutorial

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IPC-2223 — flex board design standard
- DuPont Pyralux design guide
- Multech / Sierra Circuits / Flex-Hub vendor guides
- Phil's Lab YouTube — flex PCB walkthrough
- Altium / KiCad rigid-flex tutorial videos
- IPC-6013 — flex board performance standard
- Cadence Allegro rigid-flex training
- AS9100 (aerospace) reliability requirements for flex

Thank you • Build something this week.